

Current-Loop Bandwidth Expansion Strategy for Permanent Magnet Synchronous Motor Drives

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Abstract—The dynamic response of the permanent magnet AC servo system is restricted by the bandwidth of current loop, which is the innermost loop. In digital control AC servo system, there are two major limiting factors for current-loop bandwidth, switching frequency and digital delay which means sum of A/D sampling time, algorithm execution time and PWM duty cycle update delay. The bandwidth expansion of current-loop without changing the switching frequency of power devices is necessary, as the increasing of switching frequency will raise the losses. Based on the synchronous rotating frame current decoupled control, the delay effects of current sampling and PWM duty cycle update in the permanent magnet synchronous motor drive systems were analyzed. The bandwidth expansion strategy was proposed, to achieve the stator current double sampling and PWM duty cycle double update in a carrier period. The current-loop bandwidth can be extended more than one times theoretically while the switching frequency remaining unchanged, so that the dynamic performance of permanent magnet synchronous motor drive systems was improved. Simulation and experimental results verified the theoretical analysis and effectiveness of the method.

Keywords— permanent magnet synchronous motor (PMSM); current-loop bandwidth; current sampling; PWM duty cycle update

I. INTRODUCTION

For permanent magnet synchronous motors (PMSM), torque is determined by the magnitude of the current vector as well as the rotor angle. In the applications where request fast command tracking or fast response to the load change, the servo systems dynamic performance is getting higher. The bandwidth of current-loop is important because the performance of the outer velocity-loop or position-loop is dependent on current regulation.

With the digital signal processing technology development, TMS320 series chips of TI Company are widely used as the main control core for motor drives [1, 2]. The signal processes including current sampling, current-loop regulation and space vector pulse width modulation (SVPWM) block are implemented in the digital domain. The current-loop control in the synchronous rotating reference frame has become the main form for regulating the current in AC machines due to its clear

control structure, little steady state deviation and capability to control the AC signals over a wide frequency range [3, 4].

In PM servo systems, the major factors constraining current-loop bandwidth include the PWM switching frequency and digital control loop delay. Although the switching frequency of power devices is significantly increased and some have reached the megahertz level, in the case of small and medium power servo drive system, the switching frequency is limited up to around 10 kHz subject to cost and losses [5]. Because of the limited switching frequency, the current-loop bandwidth is restricted, and is directly related to the system dynamic response capability. The digital control loop delay is comprised of the inverter dead time, current loop adjustment time, sampling time, computing time and filtering part.

Several approaches have been presented to enhance the current regulator capability under limited switching frequency. Deadbeat control can provide the fastest dynamic response speed, but the control performance is dependent on the system parameters and sensitive to changes. It is possible to reduce system stability or even instability [6]. The internal model controller has low complexity [7]. The dynamic decoupling current regulator and the complex vector current regulator are robust to the motor inductance variation and system performance does not change with synchronous frequency changes [8, 9]. And the predictive current control can improve dynamic response by using the state observer to predict the next step current [10]. Because of state variables and output variables are currents and it is the open-loop observer essentially, the observation error does not converge to zero. However, these methods still have the inherent control delay problem when using the digital signal processor (DSP) for implement of digital motor drives. There is, can not eliminate delay effects.

In this paper, the relationship among current regulator parameters, achievable bandwidth and control delay from the current sampling to the output of the PWM is established theoretically, based on dynamic model of PMSM and decoupled current control in the synchronous rotating frame. Current-loop bandwidth expansion strategy and current regulator design considerations are proposed. Simulation and experimental results verify the effectiveness of the proposed methods. The system dynamic response is improved.

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II. DYNAMIC MODEL AND CURRENT-LOOP CONTROL

A. Modeling of PMSM

The model of permanent magnet synchronous motor (PMSM) forms the basis for the control. Fig. 1 shows the equivalent circuit of PMSM. The nonlinearities due to saturation of iron and location of the stator winding in slots are neglected. The magnet flux is assumed to be sinusoidally distributed around the rotor. In the synchronous rotating reference frame, the voltage equations of PMSM become

$$u_{dqs}^r = R_s i_{dqs}^r + \frac{d\psi_{dqs}^r}{dt} + j\omega_r \psi_{dqs}^r \quad (1)$$

where $u_{dqs}^r = u_{ds}^r + ju_{qs}^r$ is the applied stator voltage vector, $i_{dqs}^r = i_{ds}^r + ji_{qs}^r$ is the stator current vector, $\psi_{dqs}^r = \psi_{ds}^r + j\psi_{qs}^r$ is the flux linkage, and R_s is the stator resistance. The flux linkage due to PM and stator currents is given by

$$\begin{bmatrix} \psi_{ds}^r \\ \psi_{qs}^r \end{bmatrix} = \begin{bmatrix} L_{sl} + L_{md} & 0 \\ 0 & L_{sl} + L_{mq} \end{bmatrix} \begin{bmatrix} i_{ds}^r \\ i_{qs}^r \end{bmatrix} + \begin{bmatrix} L_m I_f \\ 0 \end{bmatrix} \quad (2)$$

where L_{sl} is leakage inductance in the direct and quadrature direction, L_{md} the d -axis excitation inductance, and L_{mq} is the q -axis excitation inductance, respectively. For surface mounted permanent magnet machine $L_{mq} = L_{md} = L_m$.

And the stator voltage can be rewritten as

$$u_{dqs}^r = R_s i_{dqs}^r + L_s \frac{di_{dqs}^r}{dt} + j\omega_r \psi_{dqs}^r \quad (3)$$

with $L_s = L_{sl} + L_m$.

The torque in rotating coordinates becomes

$$T_{em} = p_n L_m I_f i_{qs}^r \quad (4)$$

The torque is proportional to the current in the quadrature direction.

B. Current decoupled control in the synchronous rotating reference frame

Fig. 2 shows the structure diagram of current decoupled

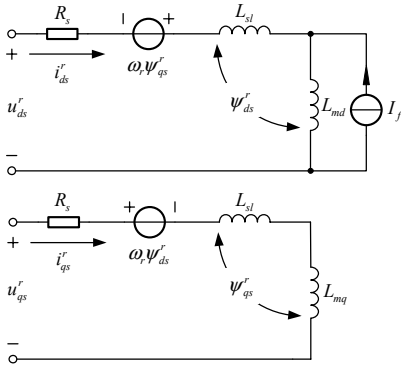


Figure 1. Equivalent circuit under synchronous rotating frame

control scheme under synchronous rotating frame. u_{dqs}^{r*} is the stator voltage reference, e_{dqs}^r is back-EMF, $u_{dqs_e}^r$ is the back-EMF compensation term for the elimination of permanent magnets impacts of the current loop, and $j\omega_r L_s$ is the cross coupling decoupled term.

The synchronous frame proportional integral (PI) current regulator has been used to cancel the plant pole $(-L_s/R_s, 0)$, with strong robustness to parameters variation. The stator voltage reference can be expressed as

$$u_{dqs}^{r*} = K_p (i_{dqs}^{r*} - i_{dqs}^r) + K_I \int (i_{dqs}^{r*} - i_{dqs}^r) dt + j\omega_r \psi_{dqs}^r \quad (5)$$

where $K_p = L_s \omega_{cb}^*$, $K_I = R_s \omega_{cb}^*$. ω_{cb}^* is the desired bandwidth. Thus, the current closed-loop transfer function is obtained

$$G_{cl}(s) = \frac{i_{dqs}^r}{i_{dqs}^{r*}} = \frac{K_p s + K_I}{L_s s^2 + (K_p + R_s)s + K_I} = \frac{\omega_{cb}^*}{s + \omega_{cb}^*} \quad (6)$$

The system dynamic response is mainly determined by the pole of the closed-loop transfer function.

When the current is well regulated, the output current i_{dqs}^r can follow the reference current i_{dqs}^{r*} . The current error is generated by the change of current reference. While (5) yields

$$R_s (i_{dqs}^r + \Delta i_{dqs}^r) + L_s \frac{\Delta i_{dqs}^r}{\Delta t} = K_p \Delta i_{dqs}^r + K_I \int \Delta i_{dqs}^r dt \quad (7)$$

where Δt is the time interval between current regulator output has finished and the next current sampling time. In the transient, the proportional control item accounts for the main part in the synchronous frame PI control. And the voltage drop in the stator resistance can be always ignored. So (7) yields

$$L_s \frac{\Delta i_{dqs}^r}{\Delta t} \approx K_p \Delta i_{dqs}^r \quad (8)$$

Substituted into the selection rule of K_p , we obtains

$$L_s \frac{\Delta i_{dqs}^r}{\Delta t} = L_s \omega_{cb}^* \Delta i_{dqs}^r \quad (9)$$

$$\omega_{cb}^* \Delta t = 1 \quad (10)$$

which, the desired current-loop bandwidth ω_{cb}^* is approximately inverse proportional to the time interval Δt .

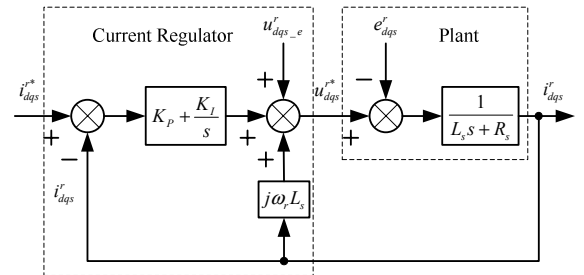


Figure 2. Structure diagram of current-loop

III. CURRENT-LOOP BANDWIDTH ANALYSIS AND EXPANSION STRATEGY

A. Current Sampling and PWM Duty Cycle Update Modes

Using TMS320F2812 as controller, the current regulator output compares with the triangular carrier and generates symmetric PWM waveform. The calculations are initiated by the interrupt events of DSP. At each interrupt, the digital controller acquires the current feedback and updates the PWM gate driving signals. The current sampling can be executed at peak or valley of triangular carrier to obtain the average value and reduce the ripple [11]. Fig. 3 shows the conventional current sampling and PWM duty cycle update time sequence. Where T_c is the carrier period, and $T_c = 1/f_c$. Single current sampling and PWM duty cycle single update (SSSU) are achieved in each carrier period.

In Fig. 3 (a), current sampling executes at peak of the triangular carrier, in the middle of the carrier period. The calculation of algorithms which are based on $(k-1)$ th current sampling executes at the beginning of k th carrier period, brings $0.5T_c$ time delay. The PWM duty cycle update executes at the beginning of $(k+1)$ th carrier period. The time delay of current regulation is 1.5 times of carrier period. Similarly, in Fig. 3 (b), where current sampling executes at the valley of the carrier, the time delay is 1 times of carrier period. As stated previously, the time delay between current sampling and PWM duty cycle update is as follows

$$T_p = \begin{cases} 1.5T_c, & \text{current sampling at peak} \\ T_c, & \text{current sampling at valley} \end{cases} \quad (11)$$

According to the current sampling at the midpoint of zero space vectors, the instantaneous sampling value will be equal to the phase current fundamental component [12]. The time delay

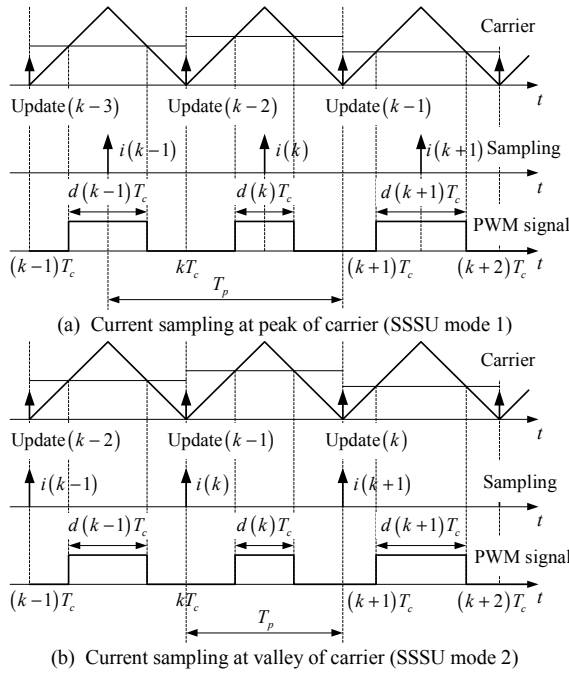


Figure 3. Time sequence of current sampling and PWM duty cycle update

in the feedback loop should be negligible. PWM inverter can generally be treated as a first-order inertia link with time constant T_{PWM} and equivalent gain K_{PWM} . Therefore, Fig. 4 shows the dynamic structure of current-loop contains time delay link.

B. Current-Loop Bandwidth Analysis

The proportional integral (PI) current regulator transfer function in Fig. 4 is

$$G_{cr}(s) = K_p + \frac{K_I}{s} = K_p \frac{T_{cr}s + 1}{T_{cr}s} \quad (12)$$

where T_{cr} is the time constant of regulator, and $T_{cr} = K_p / K_I$.

The open loop transfer function of current-loop is

$$G_{oi}(s) = \frac{K_p K_{PWM} (1/R_s) (T_{cr}s + 1)}{T_{cr}s (T_p s + 1) (T_{PWM}s + 1) (T_i s + 1)} \quad (13)$$

where T_i is the electromagnetic time constant of motor armature circuit, $T_i = L_s / R_s$. T_i is the dominant and larger time constant in the system, with $T_i \gg T_p$ and T_{PWM} . Merger the small time constants, the simplified open loop transfer function becomes

$$G_{oi}(s) = \frac{K_p K_{PWM} (1/R_s) (T_{cr}s + 1)}{T_{cr}s (T_{\Sigma i}s + 1) (T_i s + 1)} \quad (14)$$

where $T_{\Sigma i} = T_p + T_{PWM}$.

Consider the fast dynamic performance needed for current-loop control, apply current regulator parameter complying with the relation $T_{cr} = T_i$. Consequently, the closed loop transfer function of current-loop is derived as

$$G_{cli}(s) = \frac{K_p K_{PWM} (1/R_s)}{T_i T_{\Sigma i} s^2 + T_i s + K_p K_{PWM} (1/R_s)} \quad (15)$$

Further analysis of the relationship of the damping ratio ζ , open-loop cut-off frequency ω_c , -3 dB magnitude bandwidth and -45° phase shift bandwidth are as shown in Table I.

The relationship between the proportional coefficient K_p , ζ and $T_{\Sigma i}$ is shown in Fig. 5, based on the actual permanent magnet synchronous motor drive system parameters. Under the

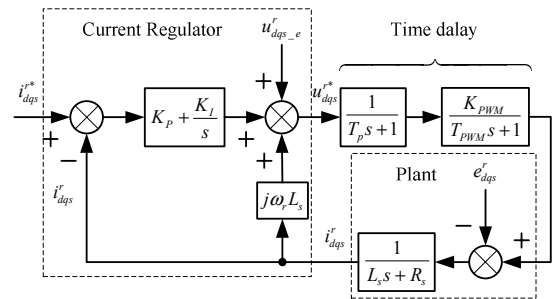


Figure 4. Structure diagram of current-loop with time delay

conditions of same damping ratio, K_p increases with $T_{\Sigma i}$ decreases. The bandwidth defined by phase shift -45° is lower than the bandwidth defined by -3 dB magnitude response and should be set as closed loop bandwidth ω_{cb} of current-loop.

Set damping ratio $\zeta = 0.707$, current regulator proportional coefficient based on the optimal 2-order property of ideal closed-loop is

$$K_p = \frac{T_i}{2K_{PWM}(1/R_s)T_{\Sigma i}} \quad (16)$$

Current-loop open loop cut-off frequency and closed-loop bandwidth are as follows

$$\omega_c = \frac{1}{T_{\Sigma i}} \sqrt{\frac{\sqrt{2}-1}{2}} \quad (17)$$

$$\omega_{cb} = \frac{\sqrt{3}-1}{2T_{\Sigma i}}, \quad f_{cb} = \frac{\omega_{cb}}{2\pi} = \frac{\sqrt{3}-1}{4\pi T_{\Sigma i}} \quad (18)$$

It can be derived that the time constant $T_{\Sigma i}$ has directly impact on the system dynamic response. Reduce $T_{\Sigma i}$ can increase the current-loop cut-off frequency and closed loop bandwidth. The conclusion is same as the approximate analysis of section II.

TABLE I. RELATIONSHIP OF DYNAMIC PERFORMANCE INDEX AND SYSTEM PARAMETERS

Parameters	Expressions
ζ	$\frac{1}{2} \sqrt{\frac{T_i}{T_{\Sigma i} K_p K_{PWM} (1/R_s)}}$
ω_c	$\sqrt{\frac{K_p K_{PWM} (1/R_s)}{T_i T_{\Sigma i}}} \sqrt{\sqrt{4\zeta^4 + 1} - 2\zeta^2}$
-3 dB BW ω_{cb1}	$\sqrt{\frac{K_p K_{PWM} (1/R_s)}{T_i T_{\Sigma i}}} \sqrt{1 - 2\zeta^2 + \sqrt{2 - 4\zeta^2 + 4\zeta^4}}$
-45° BW ω_{cb2}	$\sqrt{\frac{K_p K_{PWM} (1/R_s)}{T_i T_{\Sigma i}}} (\sqrt{1 + \zeta^2} - \zeta)$

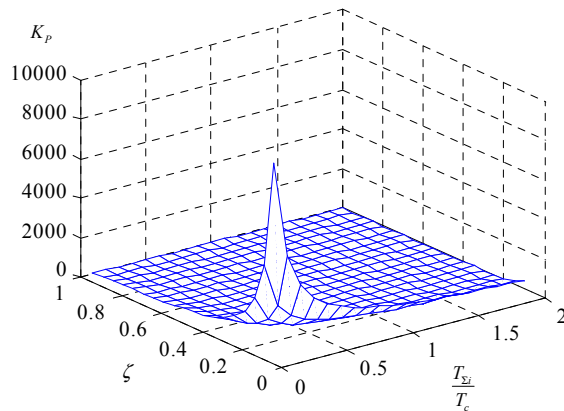


Figure 5. Relationship of K_p , ζ and $T_{\Sigma i}$

C. Current-Loop Bandwidth Expansion Strategy

According to (18), the bandwidth ω_{cb} is directly dependent on delays in the system lumped into time constant $T_{\Sigma i}$. Take SSSU modes into consideration, we obtains

$$\begin{array}{ll} \text{SSSU mode 1} & \text{SSSU mode 2} \\ \begin{cases} T_p = 1.5T_c \\ T_{PWM} = 0.5T_c \\ T_{\Sigma i} = T_p + T_{PWM} = 2T_c \end{cases} & \begin{cases} T_p = 1T_c \\ T_{PWM} = 0.5T_c \\ T_{\Sigma i} = T_p + T_{PWM} = 1.5T_c \end{cases} \end{array} \quad (19)$$

Because of T_{PWM} part is relatively fixed, it can achieve the purpose of bandwidth expansion only by reducing the T_p part of the time delay $T_{\Sigma i}$.

Fig. 6 shows the current-loop bandwidth expansion strategy by executing current sampling and PWM duty cycle update at peak and valley of the triangular carrier, two times in one carrier period. The time delay in this method becomes

$$\begin{cases} T_p = 0.5T_c \\ T_{PWM} = 0.25T_c \\ T_{\Sigma i} = T_p + T_{PWM} = 0.75T_c \end{cases} \quad (20)$$

Comparison of current control loop bandwidth with different sampling and PWM update methods is shown in Table II. Using double current sampling and PWM duty cycle double update (DSDU), the bandwidth can be 2~2.67 times higher than that of SSSU modes.

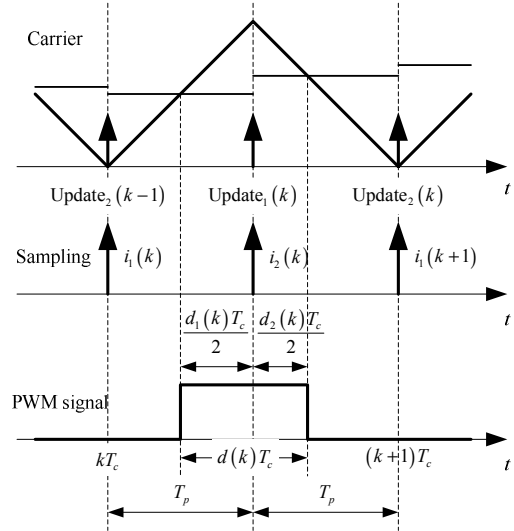


Figure 6. Strategy of double sampling and PWM duty cycle double update

TABLE II. COMPARISON OF CURRENT-LOOP BANDWIDTH WITH DIFFERENT SAMPLING AND PWM UPDATE METHODS

	Time delay $T_{\Sigma i}$	Bandwidth f_{cb}
SSSU mode 1	$2T_c$	$f_c / 34.33$
SSSU mode 2	$1.5T_c$	$f_c / 25.75$
DSDU	$0.75T_c$	$f_c / 12.86$

IV. SIMULATION AND EXPERIMENT RESULTS

A computer simulation using MATLAB/Simulink was carried out to investigate the validity of the proposed method. The parameters of the PMSM are shown in Table III. DC bus voltage is 310 V and PWM switching frequency is 10 kHz.

The open and closed loop Bode plots are shown in Fig. 7, with $\zeta = 0.707$. In DSDU mode, the open loop cut-off frequency increases to 965 Hz, corresponding to closed-loop bandwidth at -45° phase shift extends to 777 Hz. The transient response of system is improved effectively.

Fig. 8 shows the q-axis current reference and feedback waveforms. The current references are the sinusoidal waves which have frequency of 200 Hz, magnitude of 0.679 A. The DC offset in Fig. 9 (a) is 0.679 A and (b) 3.395 A. In DSDU mode, magnitude attenuation and phase lag are reduced.

TABLE III. PARAMETERS OF PMSM

Rated power	750 W
Rated torque	2.39 N·m
Rated current	4.8 A
Rated speed	3000 r/min
Maximum speed	5000 r/min
Pole pairs	4
Stator resistance	0.45 Ω
Stator inductance	3.9 mH

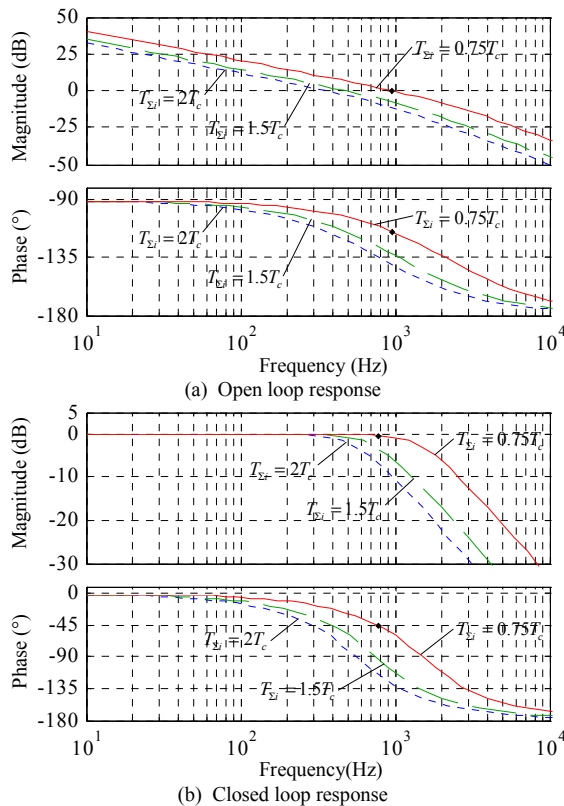
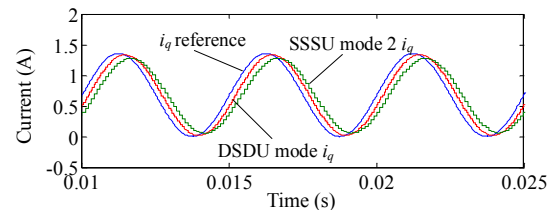


Figure 7. Frequency characteristic of the current-loop

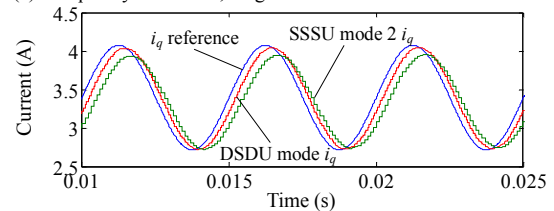
The experimental servo system is set up, consist of surface mounted PMSM, DSP control board and intelligent power module. The operation condition is the same as computer simulation to compare the results. Fig. 9 shows experimental results of q-axis current reference and actual q-axis current. The current reference frequency of 200 Hz and 333 Hz correspond to the motor rated speed and maximum speed. The DC offset are set to 0.679 A and 3.395 A, respectively. With 200 Hz and 333 Hz, 0.679 A magnitude and 0.679 A offset current reference, the phase lag of DSDU mode is 12° and 20° , while SSSU mode 2 corresponds to 25° and 41° . The phase lag of two modes with 3.395 A offset reference input are almost the same as 0.679 A offset condition. The q-axis current magnitude attenuation and phase lag increase with input frequency. Under the same given conditions, using DSDU mode has less magnitude attenuation and phase lag compared to SSSU Mode 2, and the current-loop bandwidth has been expanded.

Fig. 10 shows the step response of q-axis current under DSDU mode. The input magnitude are 1.358 A and 6.79 A. The current regulator obtains improved tracking capabilities.

Figure 11 shows the motor speed response, given the load torque alternation of idle load, half load and rated load. The real speed response curves are measured by Magtrol hysteresis dynamometer. When the loads change, the speed fluctuates greatly under SSSU mode 2. Under DSDU mode, the speed deviation is 4 r/min or less. The speed-loop response to the torque transient is improved significantly as current-loop bandwidth expanded.

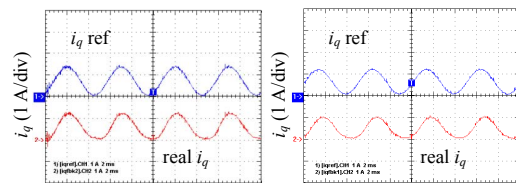


(a) Frequency of 200 Hz, magnitude of 0.679 A and 0.679 A offset



(b) Frequency of 200 Hz, magnitude of 0.679 A and 3.395 A offset

Figure 8. Simulation waveforms of current response to the current reference of 200 Hz and 750 Hz sinusoidal wave



(a) DSDU, 200 Hz, and 0.679 A offset
(b) SSSU, 200 Hz, and 0.679 A offset

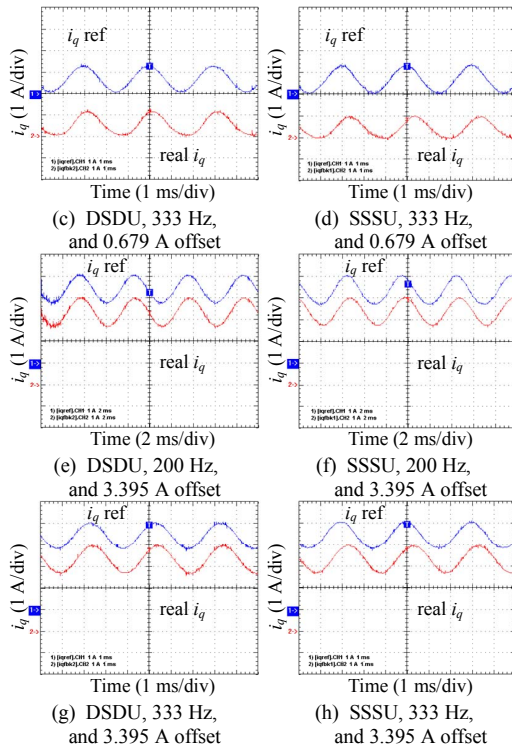


Figure 9. Experiment waveforms of current response to the current reference of 200 Hz and 333 Hz sinusoidal wave

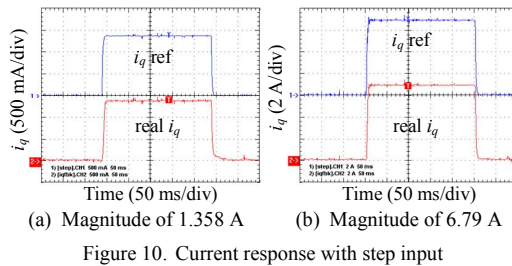
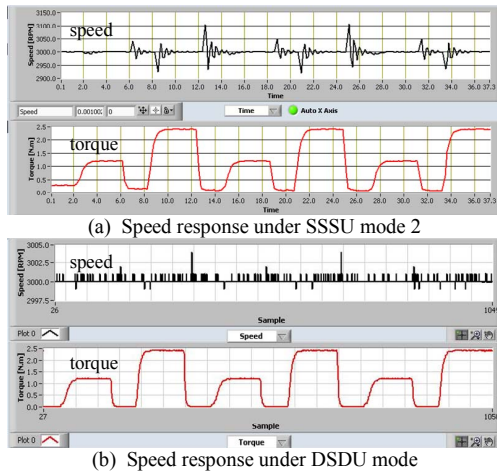


Figure 10. Current response with step input



(b) Speed response under DSDU mode

Figure 11. Speed response curves

V. CONCLUSION

In this paper, the time delay of permanent magnet AC digital control systems in different current sampling and PWM duty cycle update modes is analyzed, using current decoupled control under synchronous rotating frame. The impacts of time delay on the current-loop bandwidth are obtained. And the quantitative relationship between the two is theoretically derived. The current-loop bandwidth expansion strategy is proposed, by implementing double current sampling and double PWM duty cycle update in a carrier period. The dynamic response of system is improved while the PWM switching frequency preserves unchanged. By decreasing the time delay, the current-loop bandwidth is increased and the frequency response of speed-loop is improved. Simulation and experimental results verify the effectiveness of the method. The method proposed in this paper can provide a reference for the current-loop bandwidth expansion design in the AC motor digital control systems.

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